BOA JANG

Oculomics · Retinal Imaging Biomarkers · Longitudinal Biomarkers

I develop reliable retinal imaging biomarkers for systemic health and longitudinal disease monitoring.

balinda@snu.ac.kr · boajang97@gmail.com · Google Scholar · Homepage

EDUCATION

Seoul National University — Seoul, Korea
Ph.D. Candidate in Bioengineering — Mar. 2023 – Present

Peking University — Beijing, China
B.S. in Psychological and Cognitive Sciences — Sep. 2016 – Jul. 2020

RESEARCH EXPERIENCE

Seoul National University Hospital (SNUH) — Seoul, Korea
Medical AI Researcher — Nov. 2021 – Feb. 2023
Ph.D. Research Collaboration with SNUH — Mar. 2023 – Present

Developing AI for ophthalmic imaging in close collaboration with clinicians, spanning nAMD prognosis, retinal foundation models, and imaging biomarkers.

Korea Institute of Science and Technology (KIST) — Seoul, Korea
Research Intern, Brain Science Institute — Aug. 2020 – May 2021

Studied basal ganglia circuits and motor control through in vivo optogenetics, behavioral experiments, and fluorescence imaging.

MANUSCRIPTS UNDER REVIEW / IN REVISION

[m.5] **B. Jang** et al. Retinal Age Gap as a Biomarker of Biological Aging Reflected by Socioeconomic Status and Lifestyle Behaviors. *GeroScience* (in revision).

[m.4] **B. Jang** et al. Skeleton-Guided Progressive Test-Time Adaptation for Thin Curvilinear Structures. *AAAI 2027* (under review).

[m.3] Integrating RNFL and Fundus Photographs for Differentiation of Glaucomatous and Non-Glaucomatous Optic Neuropathy Using Deep Learning. *Scientific Reports* (in revision).

[m.2] Prediction of Postoperative Intracerebral Hemorrhage Following Brain Biopsy Using Integrated Statistical and Machine Learning Analyses. *Scientific Reports* (in revision).

[m.1] Age- and Sex-Specific Autism Screening Using Machine Learning Reduction and Conversational Agent Integration. *PLOS Digital Health* (submitted).

PEER-REVIEWED PUBLICATIONS

† Equal contribution.

[j.8] J.R. Song†, **B. Jang**†, et al. (2026). Segmentation-Guided Denoising Diffusion Probabilistic Models for Retinal OCT Speckle Noise Reduction with Enhanced Clinical Diagnostic Accuracy. *Scientific Reports* (Accepted).

[j.7] **B. Jang**, Y. Ahn, et al. (2026). Clinically Grounded Retinal Representation Learning from Minimal Supervision. *Scientific Reports.*

[j.6] **B. Jang**, Y. Shin, G. Lee, and Y.-G. Kim (2026). From Static Diagnosis to Dynamic Guidance: Evolution of Artificial Intelligence in Pediatric Neuroimaging. *Journal of Korean Neurosurgical Society.*

[j.5] J.H. Lee, S.Y. Lee, **B. Jang**, et al. (2026). OCT Biomarkers as Predictors of Treatment Interval in Neovascular Age-Related Macular Degeneration Treated with Intravitreal Aflibercept Using a Treat-and-Extend Regimen. *Scientific Reports.*

[j.4] **B. Jang**†, C.H. Lee†, et al. (2026). Artificial Intelligence-Based Prediction of First Recurrence in Neovascular Age-Related Macular Degeneration with Validation by 19 Experts. *Scientific Reports.*

[j.3] D.Y. Kim, R. Do, . . . , **B. Jang**, et al. (2025). Automated AI-Based Identification of Autism Spectrum Disorder from Home Videos. *npj Digital Medicine.*

[j.2] N.Y. Kim†, **B. Jang**†, et al. (2023). Differential Diagnosis of Pleural Effusion Using Machine Learning. *Annals of the American Thoracic Society.*

[j.1] **B. Jang**†, S.Y. Lee†, et al. (2023). Preliminary Analysis of Predicting the First Recurrence in Patients with Neovascular Age-Related Macular Degeneration Using Deep Learning. *BMC Ophthalmology.*

PRESENTATIONS

[o.7] **B. Jang**, G. Lee, J. Choi, and Y.-G. Kim. "Structure-Preserving Test-Time Adaptation for Cross-Modality Retinal Vessel Segmentation." *2026 Spring Conference of The Korean Society of Medical Informatics (KoSMI)*, Cheongju, Korea.

[po.3] **B. Jang**, K.H. Bae, and Y.-G. Kim. "Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability." *2025 Annual Conference of the Korea Society of Artificial Intelligence in Medicine (KoSAIM)*, Korea.

[o.6] **B. Jang** et al. "Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability." *MICCAI 2025 AMAI Workshop*, Daejeon, Korea. **Best Abstract Award – Honorable Mention**.

[o.5] **B. Jang** et al. "Assessment of Systemic Health Using Retinal Age Gap: Development of a Predictive Model and Clinical Applicability." *2025 Spring Conference of The Korean Society of Medical Informatics (KoSMI)*, Busan, Korea. **Best Paper Award**.

[o.4] **B. Jang** et al. "Self-Supervised Learning for Mesangial Proliferation Prediction in Digital Pathology." *The 45th Annual Meeting of the Korean Society of Nephrology (KSN 2025)*, Seoul, Korea.

[o.3] **B. Jang** and Y.-G. Kim. "Foundation Models in Healthcare: Development, Necessity and Implementation Strategies." *2025 HIMSS Global Health Conference & Exhibition (HIMSS 2025)*, Las Vegas, NV, USA.

[o.2] **B. Jang** et al. "The Impact of AI-Assistance on Predicting Recurrence in Neovascular Age-Related Macular Degeneration for Ophthalmologists." *2023 Fall Conference of The Korean Society of Medical Informatics (KoSMI)*, Bundang, Korea. **Best Paper Award**.

[po.2] **B. Jang** et al. "Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images." *10th Workshop on Ophthalmic Medical Image Analysis (OMIA-X), 2023 in International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, Vancouver, Canada.

[po.1] **B. Jang**, Y.-G. Kim, and E.K. Lee. "Predicting the First Recurrence in Patients with Neovascular AMD Using Deep Learning." *2023 American Society of Retina Specialists (ASRS)*, Seattle, WA, USA. **Poster Award Winner**.

[o.1] **B. Jang** et al. "Development and Validation of Pretrained Model for Fundoscopy-Specific using Large-Scale Images." *2023 Spring Conference of The Korean Society of Medical Informatics (KoSMI)*, Daegu, Korea. **Best Paper Award**.

AWARDS

[a.5]	AI Seoul Tech Graduate Scholarship, Seoul Future Talent Foundation	Nov. 2025
[a.4]	Hana Bank President's Award, 2024 Digital Wellness Competition	Nov. 2024
[a.3]	2nd Place, Healthcare AI Learning Platform Challenge 2023, Themes 4 & 6	Dec. 2023
[a.2]	3rd Place, Digital Pathology AI Hackathon, CODiPAI Challenge	Oct. 2023
[a.1]	SNUAIMED Excellence Scholarship	2023, 2024

TEACHING & MENTORING

[t.4]	Teaching Assistant, Korea Clinical Datathon (KCD) 2025	Oct. 2025
[t.3]	Student Ambassador, MICCAI 2025 AMAI Workshop	Sep. 2025
[t.2]	Teaching Assistant, Clinical and Healthcare AI Training Programs	2024, 2025
[t.1]	Teaching Assistant, NVIDIA MONAI Bootcamp, HCLS Summit Korea 2022	May 2022

PATENTS

[p.2] Y.-G. Kim, ..., **B. Jang**. Controllable Normal-to-Disease Fundus Translation via Factorial Guidance. Korea Patent Application No. 10-2026-0119988, filed June 2026. *(Pending)*

[p.1] H.-S. Kim, ..., Y.-G. Kim, **B. Jang**. Method and System for Predicting Information Related to Abdominal Surgery from Medical Image Data. Korea Patent No. 10-2953173, registered April 2026. *(Granted)*

REFERENCES

Prof. Jin-Wook Choi, PhD (jinchoi@snu.ac.kr)
Prof. Young-Gon Kim, PhD (younggon2.kim@gmail.com)
Prof. Hyuk Jin Choi, MD, PhD (docchoi@hanmail.net)